

Supporting Research Papers

Full reference list — ResQ-Guard / ANPR City-Wide Platform | SIH 2026 | Team Codester

1. ANPR / License Plate Detection & OCR

- **Deep Learning for License Plate Number Recognition: A Survey.** IEEE Conf. Publication. ieeexplore.ieee.org/document/10799401
- **A Survey on Deep Learning Approaches for Real-Time License Plate, Vehicle Type and Face Recognition.** IEEE Conf. Publication. ieeexplore.ieee.org/document/10127143

2. YOLOv8 Detection Performance

- **Real-Time Object Detection with YOLOv8: A Deep Learning Approach for Everyday Object Recognition.** IEEE Conf. Publication. ieeexplore.ieee.org/abstract/document/11621307
- **Real-Time Object Detection Performance of YOLOv8 Models for Self-Driving Cars in a Mixed Traffic Environment.** IEEE Conf. Publication. ieeexplore.ieee.org/document/10249521
- **Advancing Real-Time Object Detection: YOLOv8 and its Novel Approaches to Small Object Challenges.** IEEE Conf. Publication. ieeexplore.ieee.org/document/11518768

3. Multi-Camera Vehicle Tracking, Re-ID & Camera Placement

- **Tang, Z. et al. (2019).** CityFlow: A City-Scale Benchmark for Multi-Target Multi-Camera Vehicle Tracking and Re-Identification. *CVPR 2019*, pp. 8797–8806.
- **Yao, H. et al. (2022).** City-Scale Multi-Camera Vehicle Tracking Based on Space-Time-Appearance Features. *CVPR Workshops (AI City Challenge 2022, 3rd place, IDF1 0.8371)*.
- **Jain, S. et al. (2024).** Seer: A Framework for Optimizing Traffic Camera Placement and Deep Learning Inference at the Edge for Vehicle Path Reconstruction. *IEEE/ACM SEC 2024*. DOI: 10.1109/SEC62691.2024.00033
- **Ma, X. et al. (2018).** Camera Placement Based on Vehicle Traffic for Better City Security Surveillance. *IEEE Intelligent Systems*, Jul/Aug 2018.

4. Smart City Traffic Surveillance (General)

- **Visual Big Data Analytics for Traffic Monitoring in Smart City.** IEEE Conf. Publication. ieeexplore.ieee.org/document/7838263
- **Low-Power Traffic Surveillance using Multiple RGB and Event Cameras: A Survey.** IEEE Conf. Publication. ieeexplore.ieee.org/document/10293592
- **A Study of Traffic Monitoring Systems for Smart City.** IEEE Conf. Publication. ieeexplore.ieee.org/document/10430312
- **Intelligent Surveillance in Smart City Using 3D Road Monitoring.** IEEE Conf. Publication. ieeexplore.ieee.org/document/9346344
- **A Survey on Smart City Surveillance System.** IEEE Conf. Publication. ieeexplore.ieee.org/document/10146671
- **Intelligent Multi-Camera Video Surveillance System for Smart City Applications.** IEEE Conf. Publication. ieeexplore.ieee.org/document/8666579

5. Backend: Real-Time Delivery (WebSocket)

- **Pimentel, V. & Nickerson, B. G. (2012).** Communicating and Displaying Real-Time Data with WebSocket. *IEEE Internet Computing*. ieeexplore.ieee.org/document/6197172
- **Real-Time Monitoring using AJAX and WebSockets.** IEEE Conf. Publication. ieeexplore.ieee.org/document/6601579
- **Performance Analysis of Data Transmission on WebSocket for Real-time Communication.** IEEE Conf. Publication. ieeexplore.ieee.org/document/8898135
- **Real-Time Data Transmission Method Based on WebSocket Protocol for Networked Control System Laboratory.** IEEE Conf. Publication. ieeexplore.ieee.org/document/8865690

6. Backend: Spatial / Trajectory Database (PostGIS)

- **Pg-Trajectory: A PostgreSQL/PostGIS Based Data Model for Spatiotemporal Trajectories.** IEEE Conf. Publication. ieeexplore.ieee.org/document/7723677

7. ResQRoute — Emergency Green Corridor / Signal Preemption

- **Artificial Intelligence Enabled Emergency Green Corridor Mechanism for Ambulance Services in Smart City.** IEEE Conf. Publication. ieeexplore.ieee.org/document/10420235
- **Smart Ambulance System Using IoT.** IEEE Conf. Publication. ieeexplore.ieee.org/document/8336593

Coverage note: every major architecture pillar (ANPR/OCR, detection, cross-camera trajectory tracking, camera placement, general surveillance, real-time backend, spatial storage, and emergency dispatch) has at least one verified IEEE citation.